

GDP

Suchanun Oupkaew

2025-06-06

```
library('astsa')
```

```
## Warning: package 'astsa' was built under R version 4.3.3
```

```
library("forecast")
```

```
## Warning: package 'forecast' was built under R version 4.3.3
```

```
## Registered S3 method overwritten by 'quantmod':
```

```
##   method          from
```

```
## as.zoo.data.frame zoo
```

```
##
```

```
## Attaching package: 'forecast'
```

```
## The following object is masked from 'package:astsa':
```

```
##
```

```
##      gas
```

```
load("/Users/cream/Downloads/finalproject.Rdata")
tail(finalPro_data)
```

```
## # A tibble: 6 x 9
```

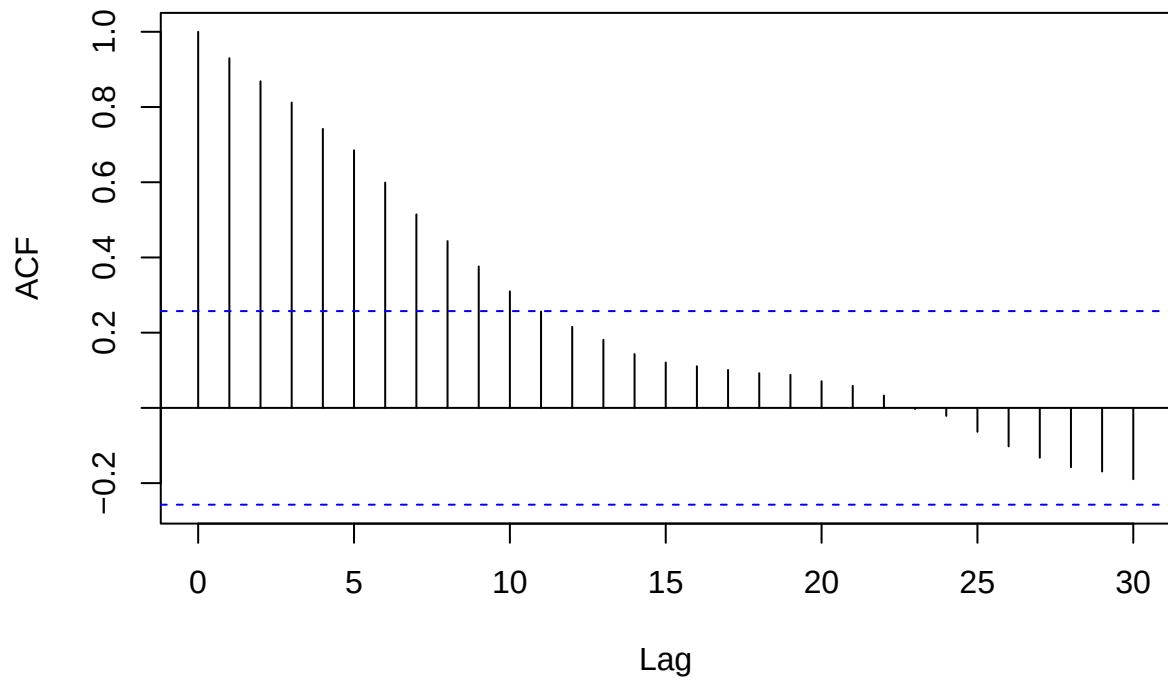
##	Country	Code	Year	GDP	Growth	CPI	Imports	Exports	Population
##	<fct>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
## 1	Central African Re~	CAF	2012	2.17e9	4.11	107.	21.8	11.6	4490416
## 2	Central African Re~	CAF	2013	1.52e9	-36.7	109.	25.0	14.5	4499653
## 3	Central African Re~	CAF	2014	1.70e9	1.04	136.	37.6	13.0	4515392
## 4	Central African Re~	CAF	2015	1.58e9	4.80	187.	34.6	12.6	4546100
## 5	Central African Re~	CAF	2016	1.76e9	4.53	NA	30.0	12.7	4594621
## 6	Central African Re~	CAF	2017	1.95e9	4.30	NA	31.8	12.5	4659080

Data Selection

```
gdp = finalPro_data$GDP
```

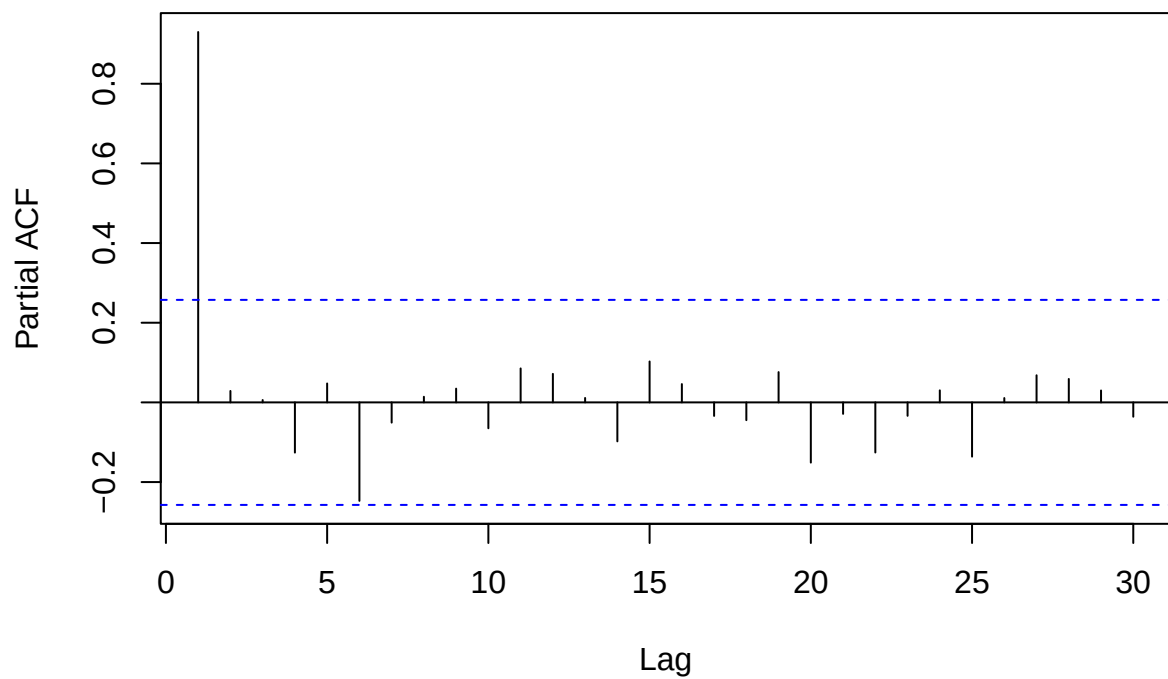
```
acf(gdp, lag.max = 30, na.action = na.pass)
```

Series gdp

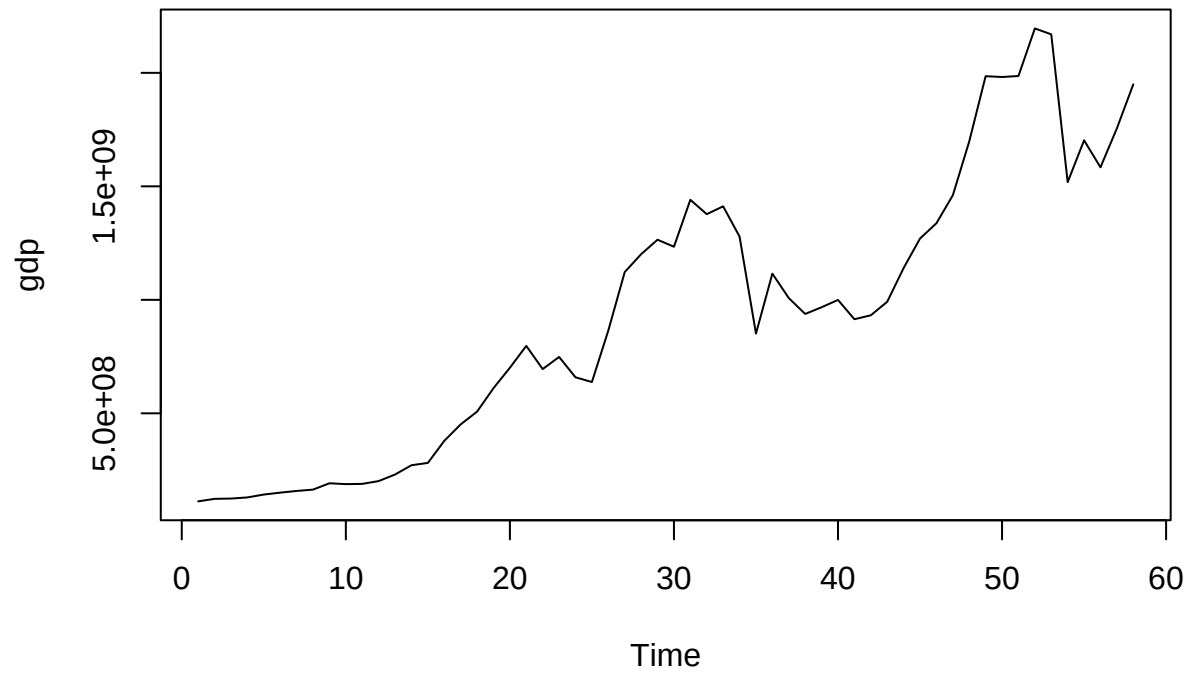


```
pacf(gdp, lag.max = 30, na.action = na.pass)
```

Series gdp

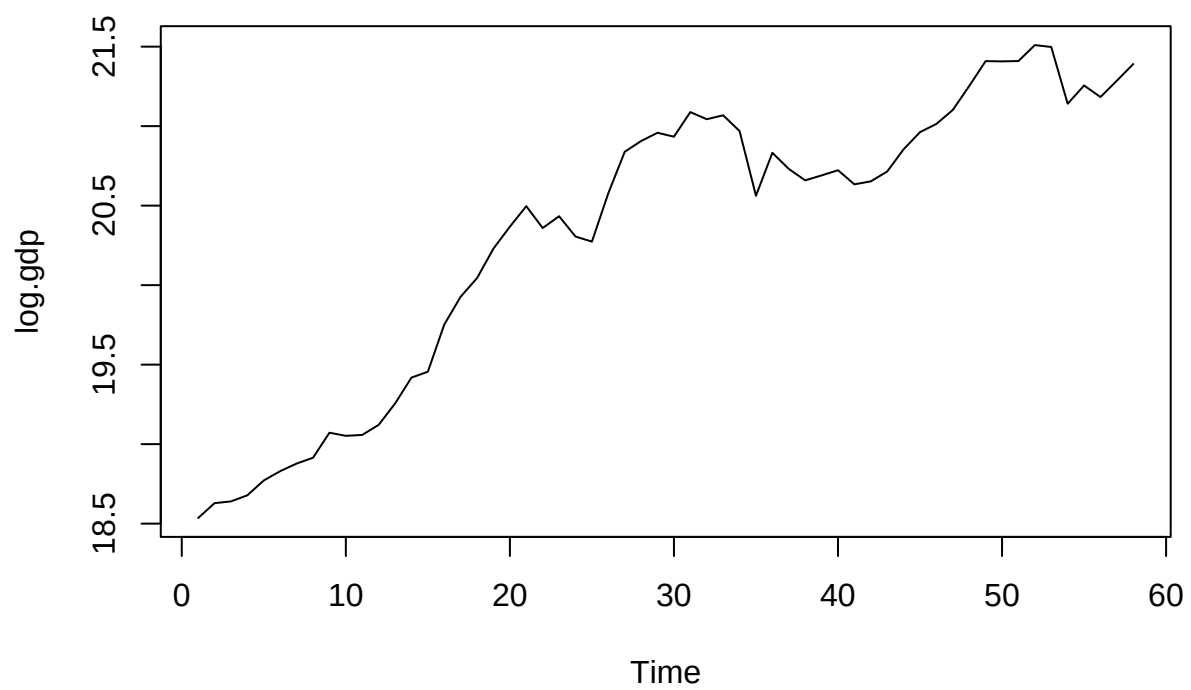


```
ts.plot(gdp)
```



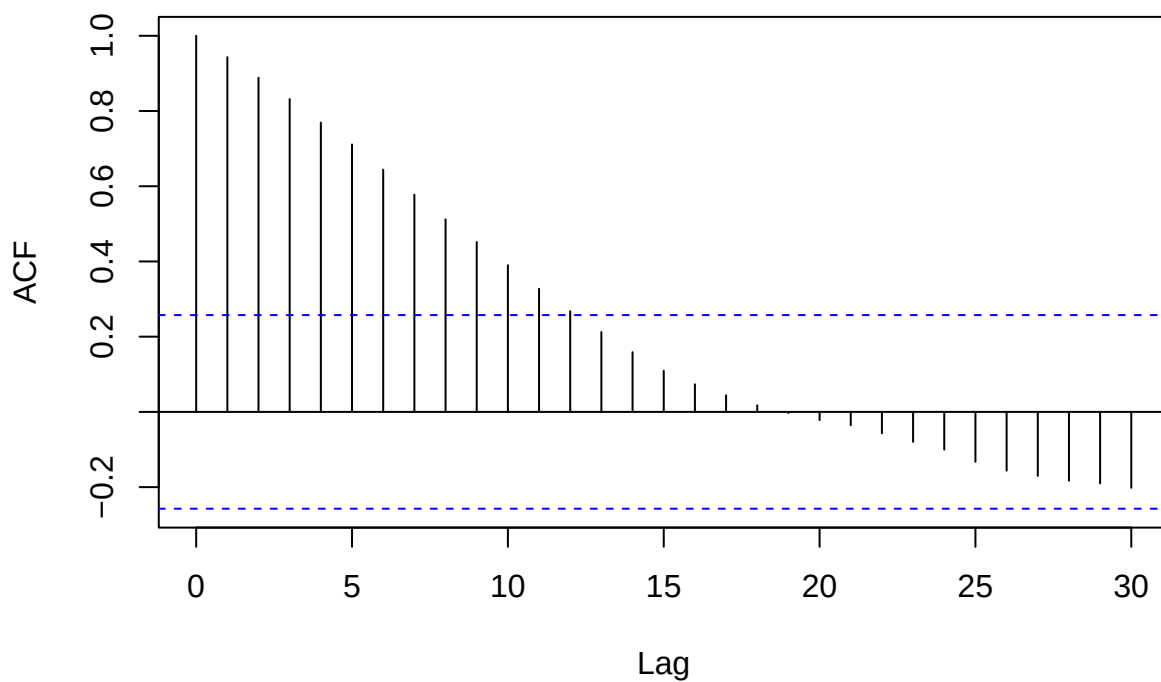
log

```
log.gdp = log(gdp)  
ts.plot(log.gdp)
```



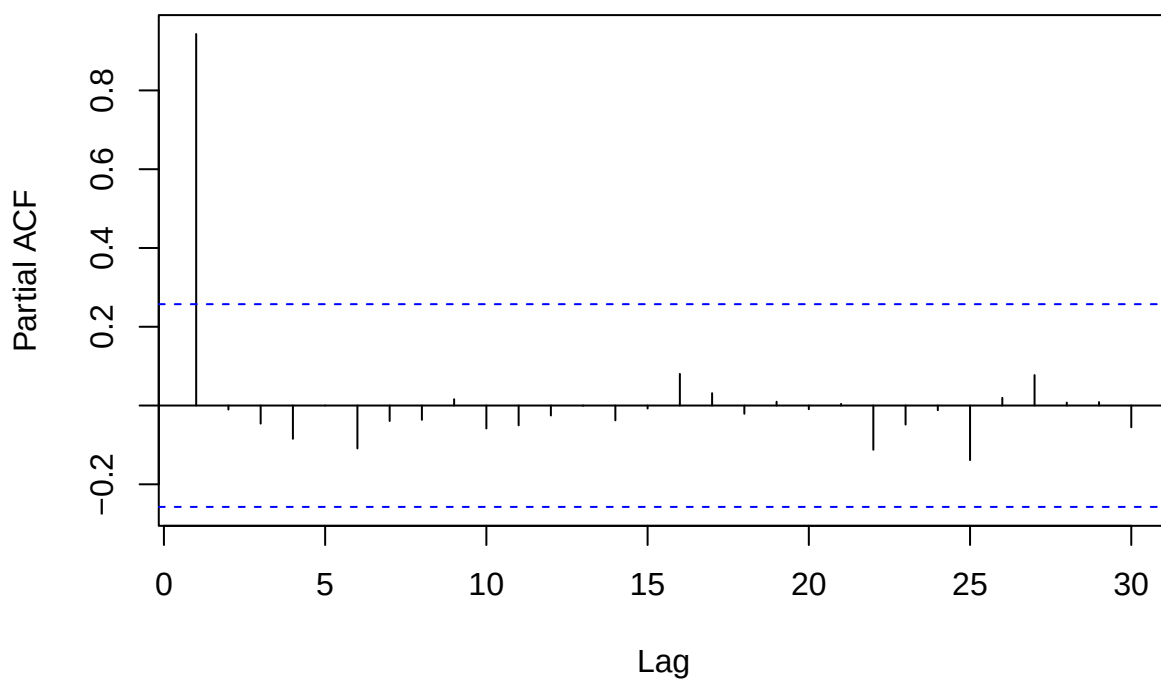
```
acf(log.gdp, lag.max = 30, na.action = na.pass)
```

Series log.gdp



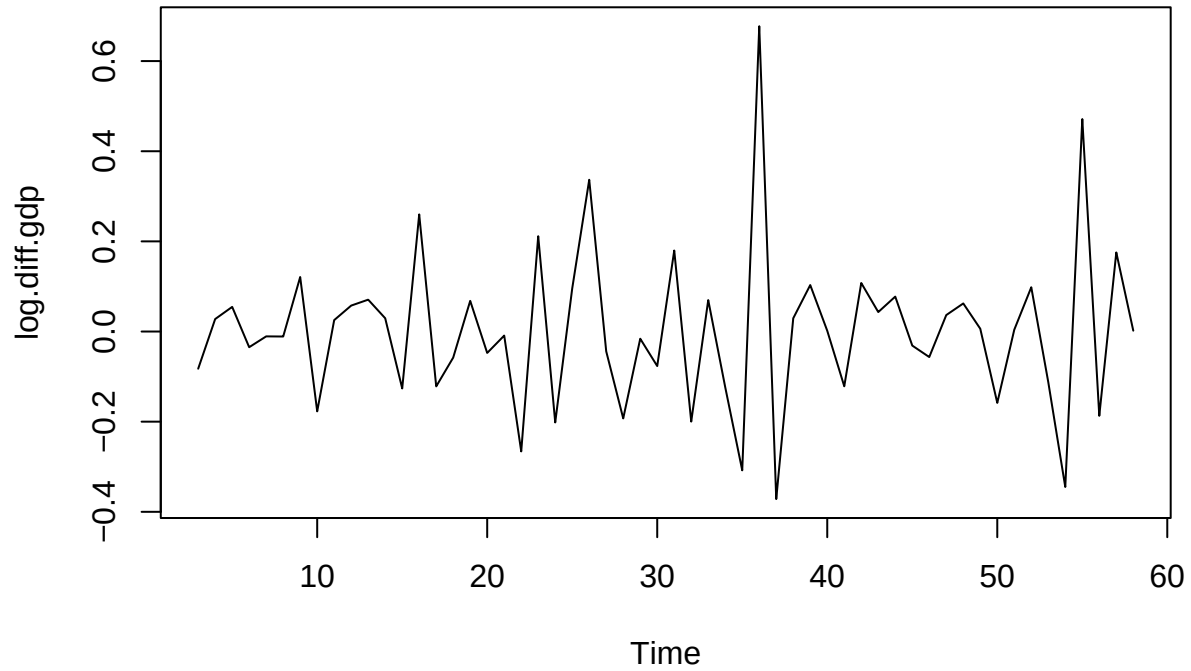
```
pacf(log.gdp, lag.max = 30, na.action = na.pass)
```

Series log.gdp



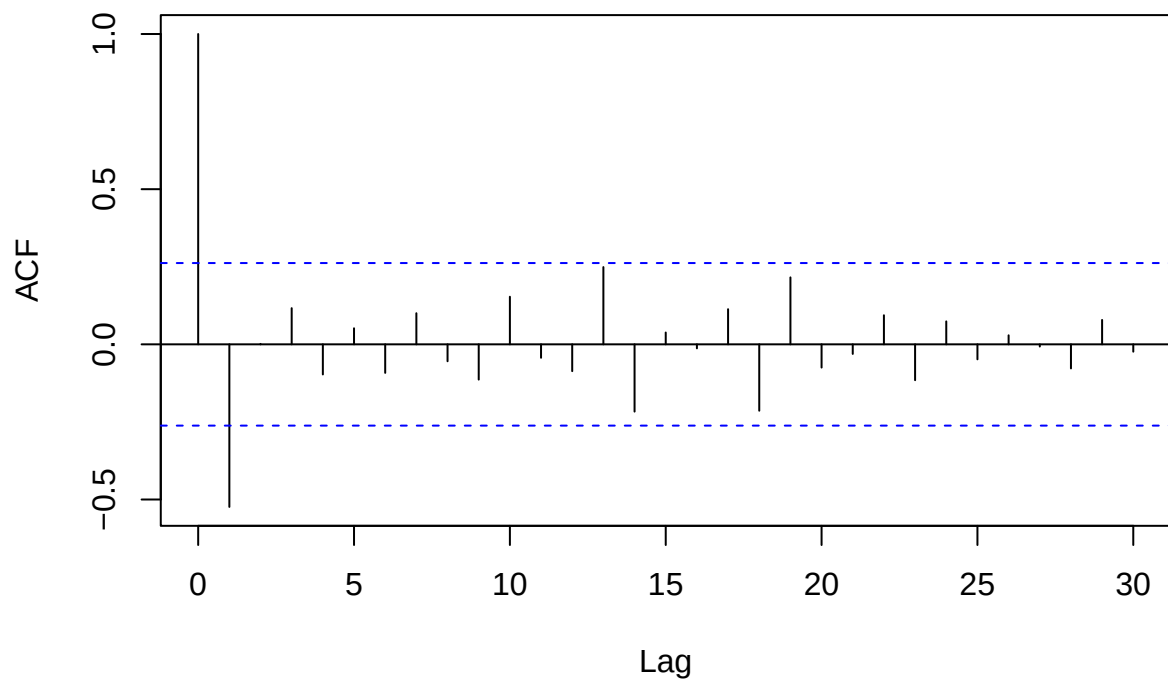
```
## log diff
log.gdp = ts(log.gdp)
log.diff.gdp = diff(log.gdp, differences = 2)

ts.plot(log.diff.gdp)
```

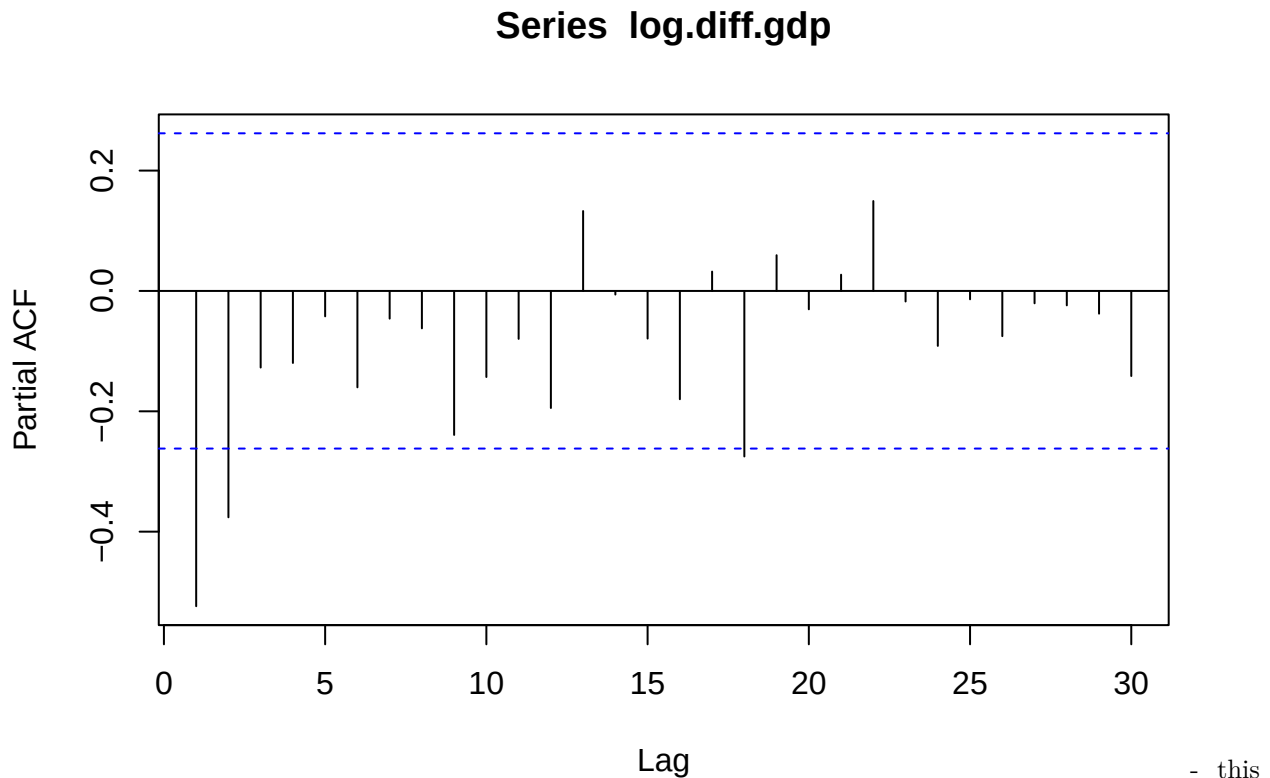


```
acf(log.diff.gdp, lag.max = 30, na.action = na.pass)
```

Series log.diff.gdp



```
pacf(log.diff.gdp, lag.max = 30, na.action = na.pass)
```



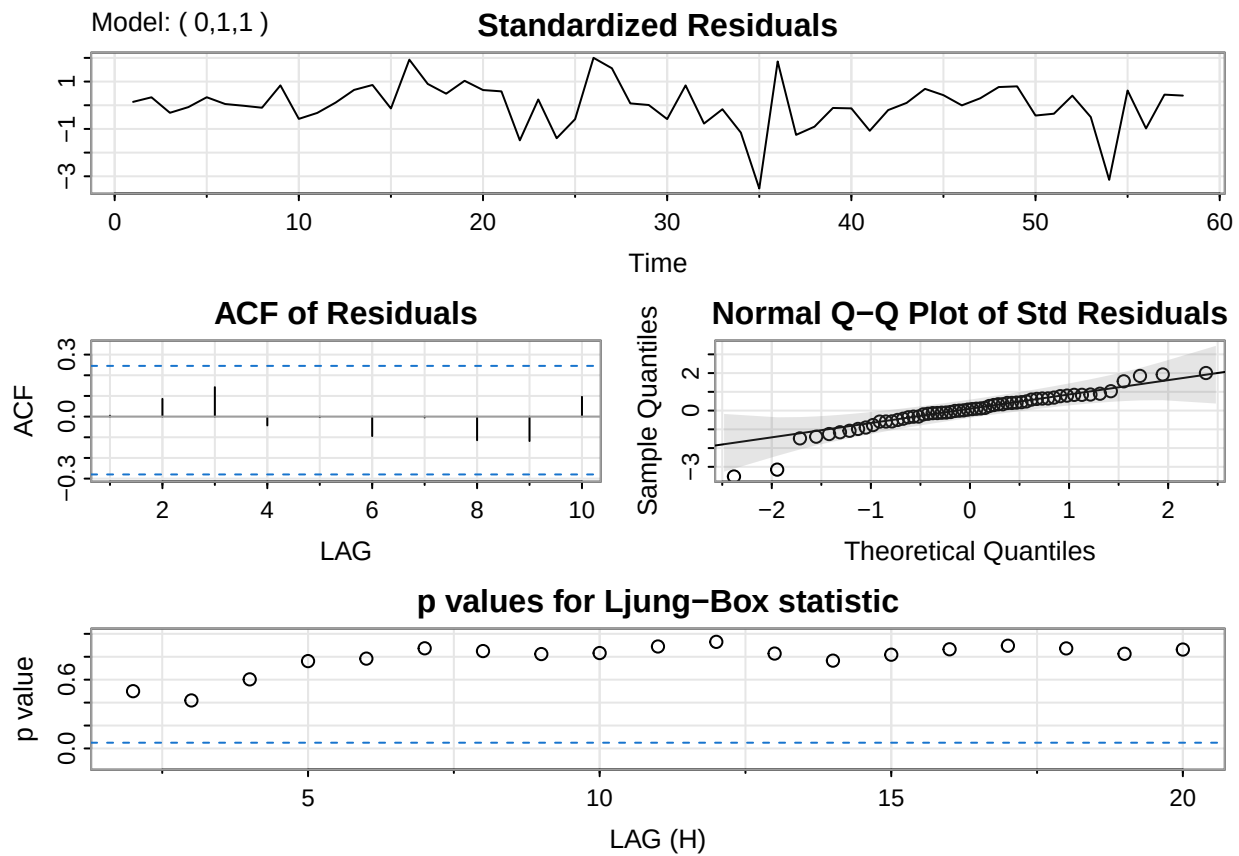
- this

seems to suggest MA(1) or MA(2) for GDP ##### MA(1) with drift = 1

```
log.gdp.ma11 = arima(log.gdp, order = c(0, 1, 1))
arima011 = sarima(log(finalPro_data$GDP), 0, 1, 1)
```

```
## initial value -2.051458
## iter 2 value -2.052302
## iter 3 value -2.052325
## iter 4 value -2.052325
## iter 4 value -2.052325
## iter 4 value -2.052325
## final value -2.052325
## converged
## initial value -2.052314
## iter 2 value -2.052314
## iter 2 value -2.052314
## iter 2 value -2.052314
## final value -2.052314
## converged
## <><><><><><><><><><><>
##
## Coefficients:
## Estimate SE t.value p.value
## ma1 0.0382 0.1228 0.3113 0.7568
## constant 0.0502 0.0177 2.8416 0.0063
##
## sigma^2 estimated as 0.01649574 on 55 degrees of freedom
```

```
##
## AIC = -1.161487 AICc = -1.157589 BIC = -1.053958
##
```



```
AIC(log.gdp.ma11)
```

```
## [1] -61.21919
```

```
Box.test(residuals(log.gdp.ma11 ), lag = 20, type = "Ljung-Box") ## res behave like white noise
```

```
##
## Box-Ljung test
##
## data: residuals(log.gdp.ma11)
## X-squared = 14.474, df = 20, p-value = 0.8057
```

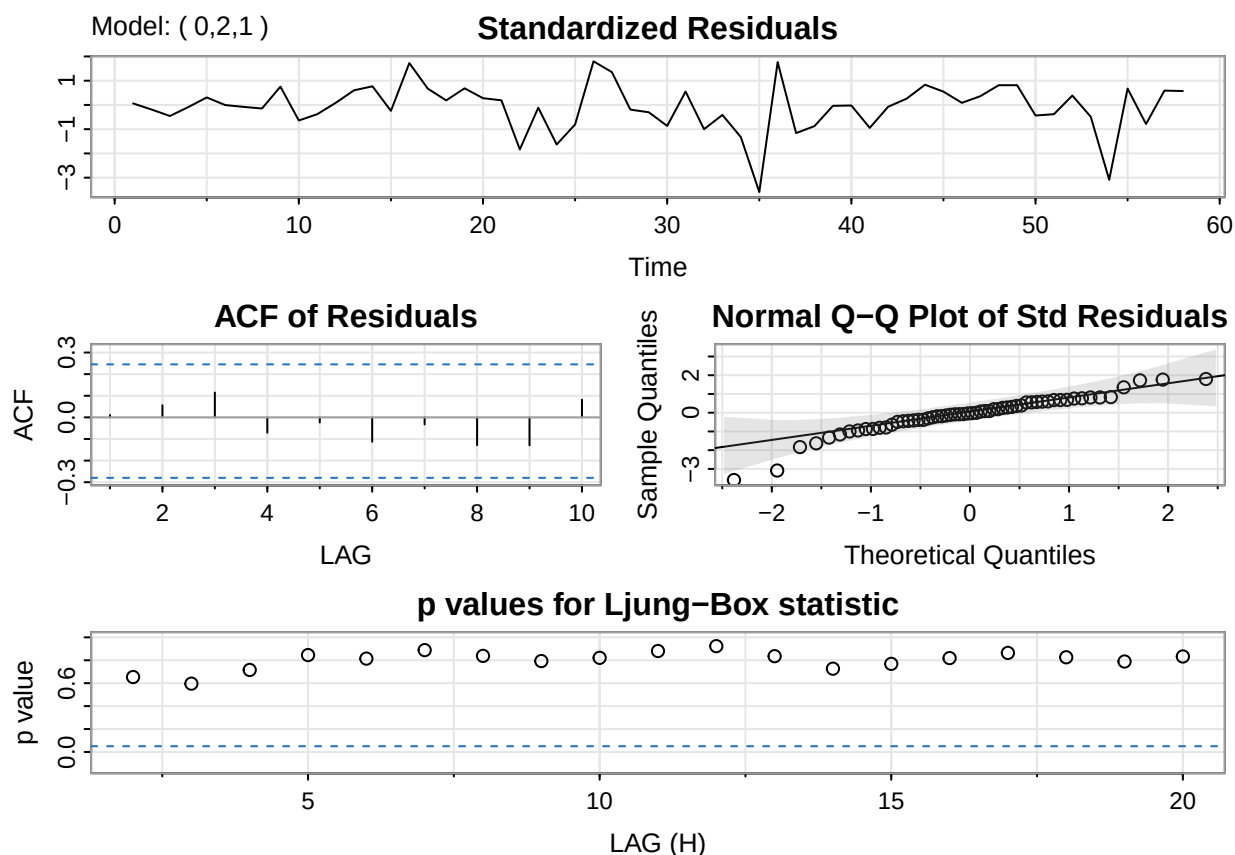
```
log.gdp.ma21 = arima(log.gdp, order = c(0, 2, 1))
arma021 = sarima(log(finalPro_data$GDP), 0, 2, 1)
```

MA(1) with drift = 2

```

## initial value -1.720403
## iter 2 value -1.935983
## iter 3 value -1.972404
## iter 4 value -1.996058
## iter 5 value -2.022799
## iter 6 value -2.026065
## iter 7 value -2.026120
## iter 8 value -2.026126
## iter 9 value -2.026135
## iter 9 value -2.026135
## iter 9 value -2.026135
## final value -2.026135
## converged
## initial value -2.009803
## iter 2 value -2.009828
## iter 3 value -2.009830
## iter 3 value -2.009830
## iter 3 value -2.009830
## final value -2.009830
## converged
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1  -0.9442 0.0704 -13.4054      0
##
## sigma^2 estimated as 0.01726135 on 55 degrees of freedom
##
## AIC = -1.110354 AICc = -1.109031 BIC = -1.03802
##

```



```
AIC(log.gdp.ma21)
```

```
## [1] -62.17981
```

```
arima021
```

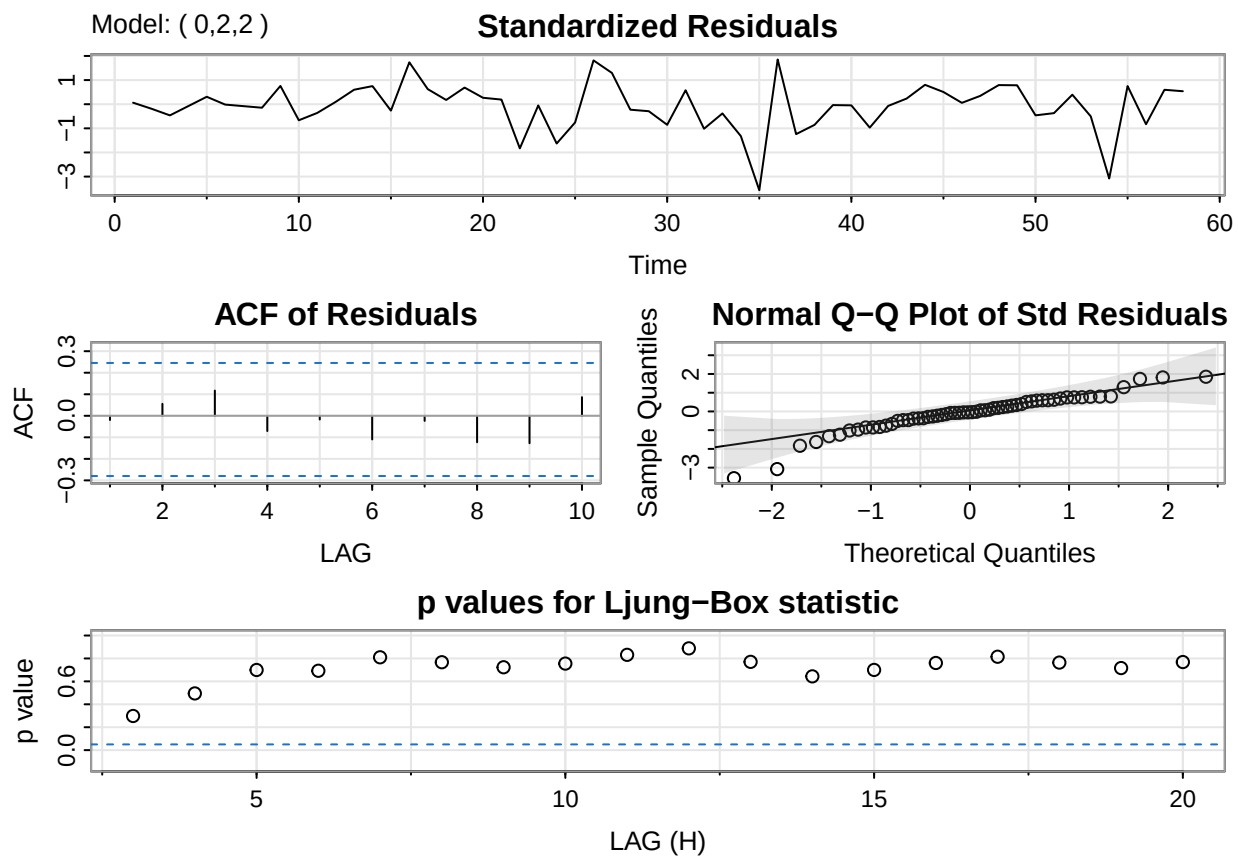
```
## $fit
##
## Call:
## arima(x = xdata, order = c(p, d, q), seasonal = list(order = c(P, D, Q), period = S),
##       include.mean = !no.constant, transform.pars = trans, fixed = fixed, optim.control = list(trace =
##       REPORT = 1, reltol = tol))
##
## Coefficients:
##          ma1
##       -0.9442
## s.e.    0.0704
##
## sigma^2 estimated as 0.01726:  log likelihood = 33.09,  aic = -62.18
##
## $degrees_of_freedom
## [1] 55
##
## $ttable
##      Estimate      SE  t.value p.value
```

```
Box.test(residuals(log.gdp.ma21 ), lag = 20, type = "Ljung-Box") ## res behave like white noise
```

```
log.gdp.ma22 = arima(log.gdp, order = c(0, 2, 2))
arima022 = sarima(log(finalPro_data$GDP), 0, 2, 2)
```

```
## initial value -1.720403
## iter 2 value -1.935860
## iter 3 value -1.979263
## iter 4 value -2.022117
## iter 5 value -2.022736
## iter 6 value -2.026141
## iter 7 value -2.026216
## iter 8 value -2.026453
## iter 9 value -2.026482
## iter 9 value -2.026482
## iter 9 value -2.026482
## final value -2.026482
## converged
## initial value -2.010324
## iter 2 value -2.010330
## iter 3 value -2.010342
## iter 4 value -2.010346
## iter 4 value -2.010346
## iter 4 value -2.010346
## final value -2.010346
## converged
## <><><><><><><><><><><>
##
## Coefficients:
##      Estimate      SE t.value p.value
## ma1  -0.9183 0.1285 -7.1446 0.0000
## ma2  -0.0303 0.1265 -0.2393 0.8118
##
## sigma^2 estimated as 0.01722835 on 54 degrees of freedom
```

```
##
## AIC = -1.075672 AICc = -1.071629 BIC = -0.9671714
##
```



```
AIC(log.gdp.ma22)
```

```
## [1] -60.23765
```

```
Box.test(residuals(log.gdp.ma22 ), lag = 20, type = "Ljung-Box") ## res behave like white noise
```

```
##
## Box-Ljung test
##
## data: residuals(log.gdp.ma22)
## X-squared = 13.38, df = 20, p-value = 0.8605
```

Forecast for GDP

```
library(forecast)
```

```
forecast(log.gdp.ma21,level=c(80,95))
```

##	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
## 59	21.42452	21.25614	21.59291	21.16700	21.68205
## 60	21.45825	21.21336	21.70314	21.08372	21.83278
## 61	21.49198	21.18371	21.80025	21.02053	21.96343
## 62	21.52571	21.16006	21.89136	20.96649	22.08493
## 63	21.55944	21.13971	21.97916	20.91752	22.20135
## 64	21.59317	21.12136	22.06498	20.87159	22.31474
## 65	21.62690	21.10422	22.14957	20.82753	22.42626
## 66	21.66062	21.08783	22.23342	20.78460	22.53664
## 67	21.69435	21.07185	22.31685	20.74232	22.64639
## 68	21.72808	21.05608	22.40009	20.70034	22.75583

```
autoplot(forecast(log.gdp.ma21))
```

